MATEMATICI 2

Curs 10

Def.: Fie $\mathcal{B}$ = sp. vect. finit peste $\mathbb{R}$

Apl.: $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ cu prop.

$(f(x,y))$

i) $\langle x, y \rangle = \langle y, x \rangle$, $\forall x, y \in V$

ii) $\langle x+y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$, $\forall x, y, z \in V$

iii) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$, $\forall \alpha \in \mathbb{R}$, $x, y \in V$

iv) $\langle x, x \rangle \geq 0$ și $\langle x, x \rangle \geq 0 \Leftrightarrow$

$\Leftrightarrow x = 0$

s.n. Produs Scalar Real

Ex.: **1)** $V = \mathbb{R}^n$, $K = \mathbb{R}$

$$\langle (x_1, \dots, x_n), (y_1, \dots, y_n) \rangle = \sum_{i=1}^{n} x_i \, y_i$$

2) $V = \mathbb{R}_n[X]$, $K = \mathbb{R}$

$$\langle P, Q \rangle = \int_{-1}^{1} P(x) \, Q(x) \, dx$$

3) $V = \mathcal{M}_{n;n\geq 2}(\mathbb{R})$, $K = \mathbb{R}$

$$\langle A, B \rangle = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} b_{ij}$$

Se pot def. prod. scalare pe sp. vect. reale care nu sunt finite.

4) $V = C_0[a,b] = \{ f : [a,b] \to \mathbb{R},\ f \text{ cont} \}$

$K = \mathbb{R}$

$$\langle f, g \rangle = \int_a^b f(x) \cdot g(x)\, dx$$

Având un prod. scalar pe V, avem prop.:

1) $\langle 0, v \rangle = 0, \quad \forall \ v \in V$

2) $\langle -v, u \rangle = - \langle v, u \rangle, \quad \forall \ v, u \in V$

Teorema/Inegalitatea lui Schwarz

$|\langle u, v\rangle| \leq \|u\| \cdot \|v\|$, unde $\|u\| = \sqrt{\langle u, u\rangle}$

$\|v\| = \sqrt{\langle v, v\rangle}$

Dem.:

Fie $f : \mathbb{R} \to \mathbb{R}$, $f(t) = \langle ut + v, ut + v \rangle \geq 0, \forall t \in \mathbb{R}$

$u, v \in V$ fixați, $u \neq 0$

$f(t) = t^2 \langle u, u \rangle + t \langle u, v \rangle + t \langle u, v \rangle + \langle v, v \rangle$

$= t^2 \underbrace{\langle u, u \rangle}_{\|u\|^2 > 0} + 2t \langle u, v \rangle + \underbrace{\langle v, v \rangle}_{\|v\|^2 \geq 0} \geq 0$

$\forall t \in \mathbb{R} \iff \Delta \leq 0 \iff 4 \langle u, v \rangle^2 - 4 \langle u, u \rangle \langle v, v \rangle \leq 0 \,/ : 4$

$\langle u, v \rangle^2 \leq \langle u, u \rangle \langle v, v \rangle$

$|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$

Obs.: Din Schwarz $\Rightarrow$ $-1 \leq \frac{\langle u,v\rangle}{\|u\|\cdot\|v\|} \leq 1$

Notăm $\cos(u,v) = \frac{\langle u,v\rangle}{\|u\|\cdot\|v\|}$

Ex.: Să se dem ~~$\forall n \in \mathbb{N}^*$~~ T. Pitagora generalizată.

pe un sp. vectorial înzestrat cu prod. scalar

Def.: Dc. V = sp. vect. cu prod. scalar real (/ complex)

$v \perp u \iff \langle v, u \rangle = 0$

$\perp$ = ortogonal

Def.: Fie V = sp. vect. finit peste $\mathbb{C}$

Apl.: $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$ cu prop.

i) $\langle x, y \rangle = \overline{\langle y, x \rangle}$, $\forall x, y \in V$

ii) $\langle x+y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$, $\forall x, y \in V$

iii) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$, $\forall \alpha \in \mathbb{C}$, $x, y \in V$

iv) $\langle x, x \rangle \geq 0$ și $\langle x, x \rangle = 0 \Leftrightarrow x = 0$

s.n. Produs Scalar Complex

Ex.: $V = \mathbb{C}^n$, $K = \mathbb{C}$

$$\langle (x_1, \dots, x_n), (y_1, \dots, y_n) \rangle = \sum_{i=1}^{n} x_i \overline{y_i}$$

Analog prod. scalar real, avem ineg. lui Schwarz

și $\| v \|$ (normă dintr-un vector).

Def.: Fie V = sp. vect. finit peste $\mathbb{C}$

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iii) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$, $\forall \alpha \in \mathbb{C}$, $x, y \in V$

iv) $\langle x, x \rangle \geq 0$ și $\langle x, x \rangle = 0 \Leftrightarrow x = 0$

s.n. Normă

Se poate def. o normă la fel și pe un sp. vect. COMPLEX

Ex.: Dc. V = sp. vectorial cu prod. scalar real

atunci

$\|\cdot\| : V \to \mathbb{R}$; $\|v\| = \sqrt{\langle v, v \rangle}$ este normă

$\|v\| = \sqrt{\langle v, v \rangle} \geq 0$

$\|v\| = 0 \Leftrightarrow \sqrt{\langle v, v \rangle} = 0 \Leftrightarrow v = 0$

$$\|v+u\| = \sqrt{\langle v+u, v+u\rangle} = \sqrt{\langle v,v\rangle + \langle v,u\rangle + \langle u,v\rangle + \langle u,u\rangle} =$$

$$= \sqrt{\|v\|^2 + 2\langle u,v\rangle + \|u\|^2} \leq$$

$$\leq \sqrt{\|v\|^2 + 2\|u\|\cdot\|v\| + \|u\|^2} =$$

$$= \sqrt{(\|v\| + \|u\|)^2} = \|v\| + \|u\|$$

În cazul în care prod. scalar este COMPLEX, avem:

$$|\langle v,u\rangle + \langle u,v\rangle| = |\underbrace{\langle v,u\rangle + \langle \overline{v},\overline{u}\rangle}_{\in \mathbb{R}}| \leq$$

$$\leq |\langle v,u\rangle| + |\langle \overline{v},\overline{u}\rangle| \leq \|v\|\cdot\|u\| + \|v\|\cdot\|u\|$$

$$\|v+u\| = \sqrt{\langle v,v\rangle + \langle u,u\rangle + \langle u,v\rangle + \langle u,u\rangle} \leq$$

$$\leq \sqrt{\|v\|^2 + 2\|v\|\cdot\|u\| + \|u\|^2} = \|v\| + \|u\|$$

$$\|\alpha v\| = \sqrt{\langle \alpha v, \alpha v\rangle} = \sqrt{\alpha \cdot \bar{\alpha} \cdot \langle v, v\rangle} =$$

$$= \sqrt{|\alpha|^2 \langle v, v\rangle} = |\alpha| \sqrt{\langle v, v\rangle} = |\alpha| \cdot \|v\| \Rightarrow \|\cdot\| \text{ norma}$$

$$\langle v, \alpha v\rangle = \overline{\langle \alpha v, v\rangle} = \overline{\alpha \langle v, \bar{v}\rangle} = \bar{\alpha} \langle \bar{v}, \bar{v}\rangle = \bar{\alpha} \langle v, v\rangle$$

Propr.: Fie V sp. vect. finit real sau complex înzestrat cu prod. scalar.

Dc. $\|v\| = \sqrt{\langle v, v \rangle}$, atunci $d : V \times V \to \mathbb{R}$ ($\mathbb{C}$)

$d(v, u) = \|v - u\|$ = distanța

și (V, d) sp. metric.

Def.: Un sp. vect. înzestrat cu prod. scalar care, împreună cu distanța dată de norma prod. de la prod. scalar, este SP. METRIC COMPLET

s.n. Sp. Hilbert

Def.: Un sp. vectorial finit înzestrat cu prod. scalar

s.n. Sp. Euclidian

Def.: Fie V sp. vect. euclidian.

Mult. de vectori $\{v_1, \dots, v_n\}$ este ortogonală $\Leftrightarrow \langle v_i, v_j \rangle = 0$, $\forall\, i \neq j,\ i, j \in \overline{1,n}$

Propr.: $\forall$ mult. ortogonală de vectori (mult. de vectori ortogonali NENULI) e liniar indep..

Dem.: $a_1 v_1 + \dots + a_n v_n = 0$

$\langle a_1 v_1 + \dots + a_n v_n , v_i \rangle = 0 , v_i \rangle = 0, \quad i = \overline{1,n}$

$\sum_{j=1}^{n} a_i \underbrace{\langle v_j, v_i \rangle}_{= 0 \text{ pt. } j \neq i} = 0$

$a_i \underbrace{\langle v_i, v_i \rangle}_{\neq 0} = 0 \Rightarrow a_i = 0, \quad i = \overline{1,n}$

$\forall v_i \neq 0$

Dc. $\dim_K V = n \Rightarrow \{v_1, \dots, v_n\}$ = ortogonali $\Leftrightarrow$

$\Leftrightarrow \{v_1, \dots, v_n\}$ = bază ortogonală

Def.: Dc. V = sp. vect. finit înzestrat cu prod. scalar

$\dim V = n$

atunci $B = \{v_1, \dots, v_n\}$ bază este

bază ortonormală $\Leftrightarrow \langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$

(vectorii sunt ortogonali și au fiecare norma 1)

Ex.: **i)** $V = \mathbb{R}^n$, $K = \mathbb{R}$

$$\langle (x_1, \dots, x_n), (y_1, \dots, y_n) \rangle = \sum_{i=1}^{n} x_i y_i$$

$B = \{ e_1 = (1, 0, \dots, 0), \dots, e_n = (0, \dots, 0, 1) \} =$

= bază ortonormală

ii) $V = V_3$, $K = \mathbb{R}$

$\langle \bar{v}, \bar{u} \rangle = \bar{v}\,\bar{u}$

$B = \{\bar{i}, \bar{j}, \bar{k}\}$ bază ortonormală

iii) $V = M_{n,m}(\mathbb{R})$, $\langle A, B \rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} b_{ij}$

$B = \{ E_{ij} \mid i = \overline{1,n},\ j = \overline{1,m},\ E_{ij} = (e_{kl})_{\substack{1 \le k \le n \\ 1 \le l \le m}} \}$

$$e_{kl} = \begin{cases} 1, & \text{pt. } (k,l) = (i,j) \\ 0, & \text{pt. în rest} \end{cases}$$